

Frontend Development Internship

Task 2: Groovebox Music Player

Project Documentation

Project Name	Groovebox Music Player
Project Type	Frontend Development Project
Live Website	https://groovebox-music-player.netlify.app/
GitHub Repository	https://github.com/04nandini0/groovebox-music-player

1. Project Overview

Groovebox is a frontend UI recreation of a music streaming application, built as a practical exercise in replicating a modern, real-world interface using HTML, CSS, and JavaScript. The live application itself identifies as a "UI recreation project built by Nandini Sharma for learning purposes," and its interface — a dark-themed layout with a left sidebar, a scrollable content area of playlists and recommendations, and a persistent bottom player bar — is directly modelled on the visual language of mainstream music streaming platforms.

The purpose of building this project was to practice recreating a complex, familiar interface from a visual and interaction standpoint: laying out a multi-panel application (navigation sidebar, content grid, player bar), and wiring up JavaScript-driven UI behaviour such as play/pause state, progress tracking, and track switching.

The user experience goal was to closely mirror the look, feel, and navigation flow of a professional music streaming product, so that a visitor interacting with Groovebox — browsing playlists, selecting a track, and using the player controls — encounters an interface that feels familiar and polished, despite being a frontend-only recreation.

2. Project Objective

- Practice frontend development by recreating a multi-section, real-world application layout from scratch.
- Build an interactive music player interface, including a functioning play/pause control, progress bar, and time display.
- Understand and apply UI design principles used in modern streaming platforms — dark theming, card-based content grids, and persistent player bars.
- Implement frontend functionality using vanilla JavaScript for state changes such as toggling playback and updating the currently selected track.
- Ensure the interface adapts responsively for smaller/mobile screen widths.
- Deploy the finished application to a live hosting platform and publish the source code on GitHub.

3. Technologies Used

The application is built entirely with core frontend web technologies, consistent with the live site's own description of itself as an HTML/CSS/JavaScript UI recreation project.

Technology	Purpose
HTML5	Semantic structure for the player layout — sidebar navigation, main content area, and the bottom player bar.
CSS3	Dark, Spotify-inspired visual styling, sidebar and card layouts, and responsive layout adjustments for smaller screens.
JavaScript (Vanilla)	UI interactivity — play/pause toggle, progress bar and time-display updates, track selection, and sidebar/menu behaviour.
Netlify	Deployment and hosting of the live web application.

4. Features Implemented

4.1 Sidebar Navigation

A left-hand sidebar provides primary navigation with Home, Search, and Your Library links, along with a Create Playlist option and a dedicated Liked Songs shortcut. Below these, a scrollable list of user playlists is displayed (e.g., Road Trip Anthems, Study Beats, 90s Throwback, Workout Mix, Rainy Day, Coffee Shop Jazz, Late Night Vibes).

4.2 Top Bar

The top bar includes back/forward navigation arrows, a search input field ("What do you want to listen to?"), and a user profile indicator displaying the account name (Nandini) with an avatar icon.

4.3 Content Sections

The main content area presents music in curated, card-based rows, including "Recently played," "Made for you," and "Trending music." Each card displays cover art, a track/playlist title, and an artist or descriptive subtitle (e.g., Discover Weekly — "Fresh picks based on what you play").

4.4 Music Playback Controls

A persistent player bar at the bottom of the screen displays the currently selected track's cover art, title, and artist, along with a like (heart) icon. Playback controls include shuffle, previous track, a central play/pause button, next track, and repeat icons. Selecting and playing a track updates the play button to a pause icon and the currently displayed track information.

4.5 Progress Bar and Time Display

The player bar includes a horizontal progress bar with elapsed and total time shown on either side (e.g., 0:03 / 0:12). The progress bar fill and elapsed time update as a track plays.

4.6 Volume Control

A volume icon paired with a horizontal slider is available at the right end of the player bar for adjusting playback volume.

4.7 Responsive Interface

On narrower screen widths, the layout adapts: the sidebar collapses into a stacked mobile view accessible via a menu icon, and the content grid reflows into a single column, while the player bar remains fixed at the bottom of the screen.

5. UI/UX Design

5.1 Layout Structure

The interface follows a classic three-zone music player layout: a fixed sidebar on the left for navigation and playlists, a scrollable main content panel in the centre for browsing music, and a fixed player bar anchored to the bottom of the viewport that remains visible regardless of scroll position.

5.2 Visual Design Choices

- A dark background theme is used throughout, consistent with mainstream music streaming applications, reducing visual strain and putting focus on the colourful album-art cards.
- Album/playlist artwork is represented using colourful geometric shapes (circles, squares, triangles, ovals) on solid colour backgrounds, giving each card a distinct, recognizable identity.
- A green accent colour is used for interactive highlights (e.g., the logo icon and the progress bar fill), echoing familiar streaming-platform branding.
- Consistent card sizing and spacing across the "Recently played," "Made for you," and "Trending music" rows keep the browsing experience visually organized.

5.3 User Interaction Flow

A visitor lands on the Home view, browses playlists and recommended content via scrollable card rows, and selects a track to begin playback. Once a track is selected, the bottom player bar updates to reflect the current song, and the visitor can control playback (play/pause, next/previous), like the track, and adjust volume — all without leaving the current view, since the player bar persists across the interface.

5.4 Responsive Behavior

At mobile widths, the sidebar transforms into a compact stacked menu (accessible via a hamburger-style icon) and the multi-column content grid collapses into a single vertical column, while the bottom player bar continues to remain fixed and accessible.

6. Development Process

6.1 Planning

The project began by identifying the core structural zones common to music streaming interfaces — sidebar navigation, a browsable content area, and a persistent player bar — and planning how each would be implemented and connected.

6.2 Designing the Interface

The visual design was planned around a dark theme with card-based content rows and a green accent colour, closely mirroring the conventions of well-known streaming platforms while using original placeholder artwork (geometric shapes) rather than real album art.

6.3 Implementing Functionality

HTML was used to structure the sidebar, content grid, and player bar; CSS handled the dark theming, card layouts, and responsive breakpoints; and JavaScript was used to implement interactive behaviour such as toggling play/pause state, updating the progress bar and time display, and switching the currently displayed track.

6.4 Testing

The interface was manually tested by navigating the sidebar links, selecting different tracks from the content cards, operating the play/pause and next/previous controls, and resizing the browser window to verify the layout adapted correctly at smaller screen widths.

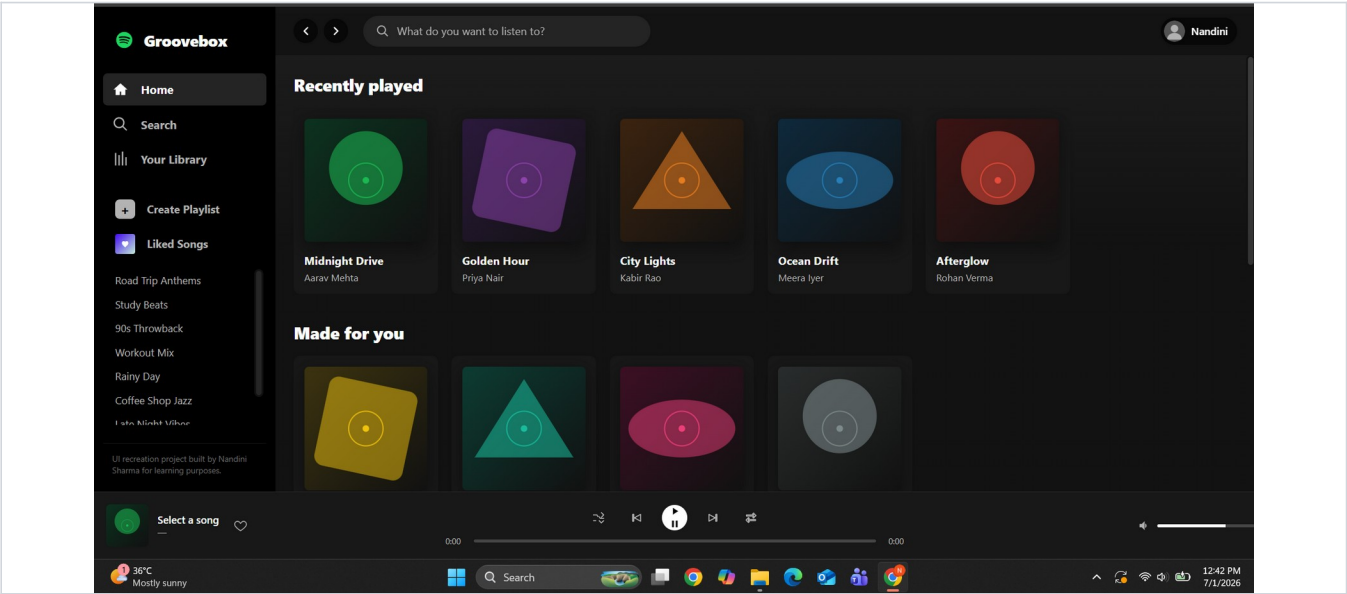
6.5 Deployment

The completed project was pushed to a GitHub repository for version control and source code access, and deployed to Netlify for live public hosting.

7. Screenshots Documentation

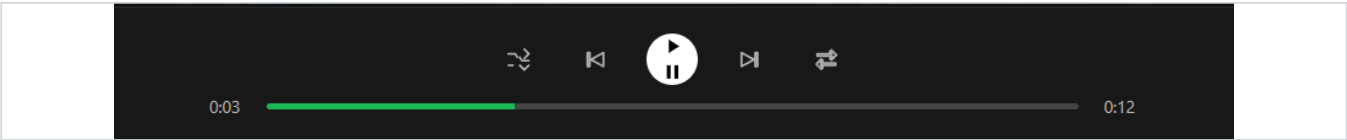
The following screenshots were captured directly from the live, deployed Groovebox application.

Figure 1: Main Music Player Interface



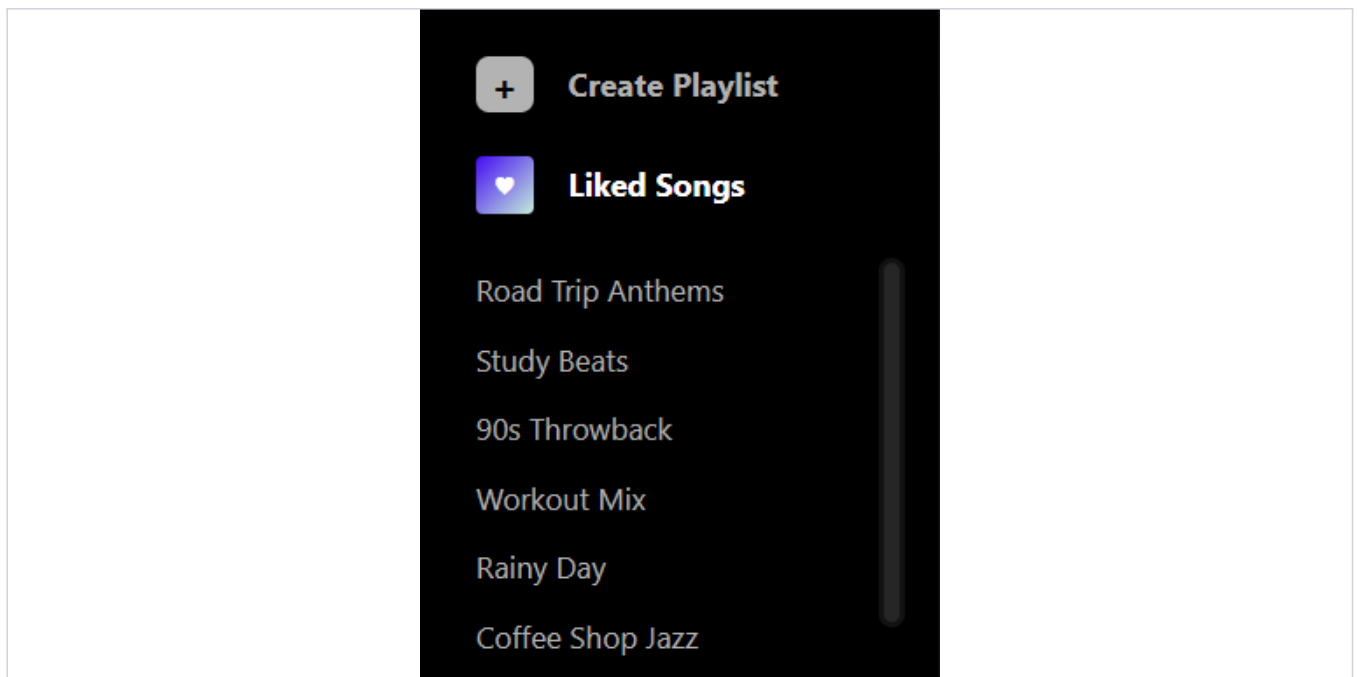
Shows the primary Home view, including the sidebar navigation, top search bar with user profile, the "Recently played" and "Made for you" content rows, and the bottom player bar with playback controls.

Figure 2: Player Controls Section



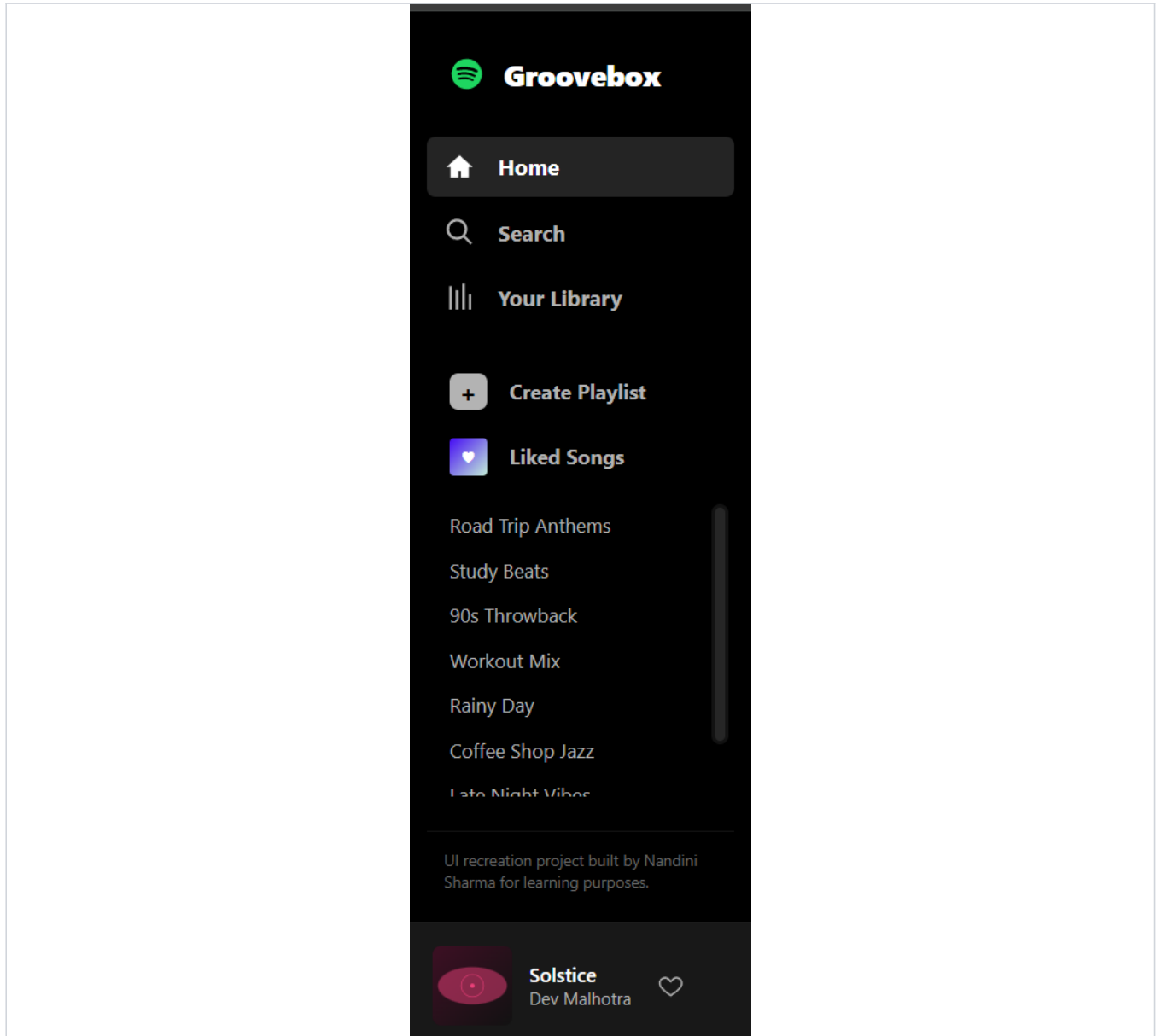
A close-up of the playback control bar showing shuffle, previous track, play/pause, next track, and repeat icons, along with the progress bar and elapsed/total time display (0:03 / 0:12) during active playback.

Figure 3: Playlist / Library Section



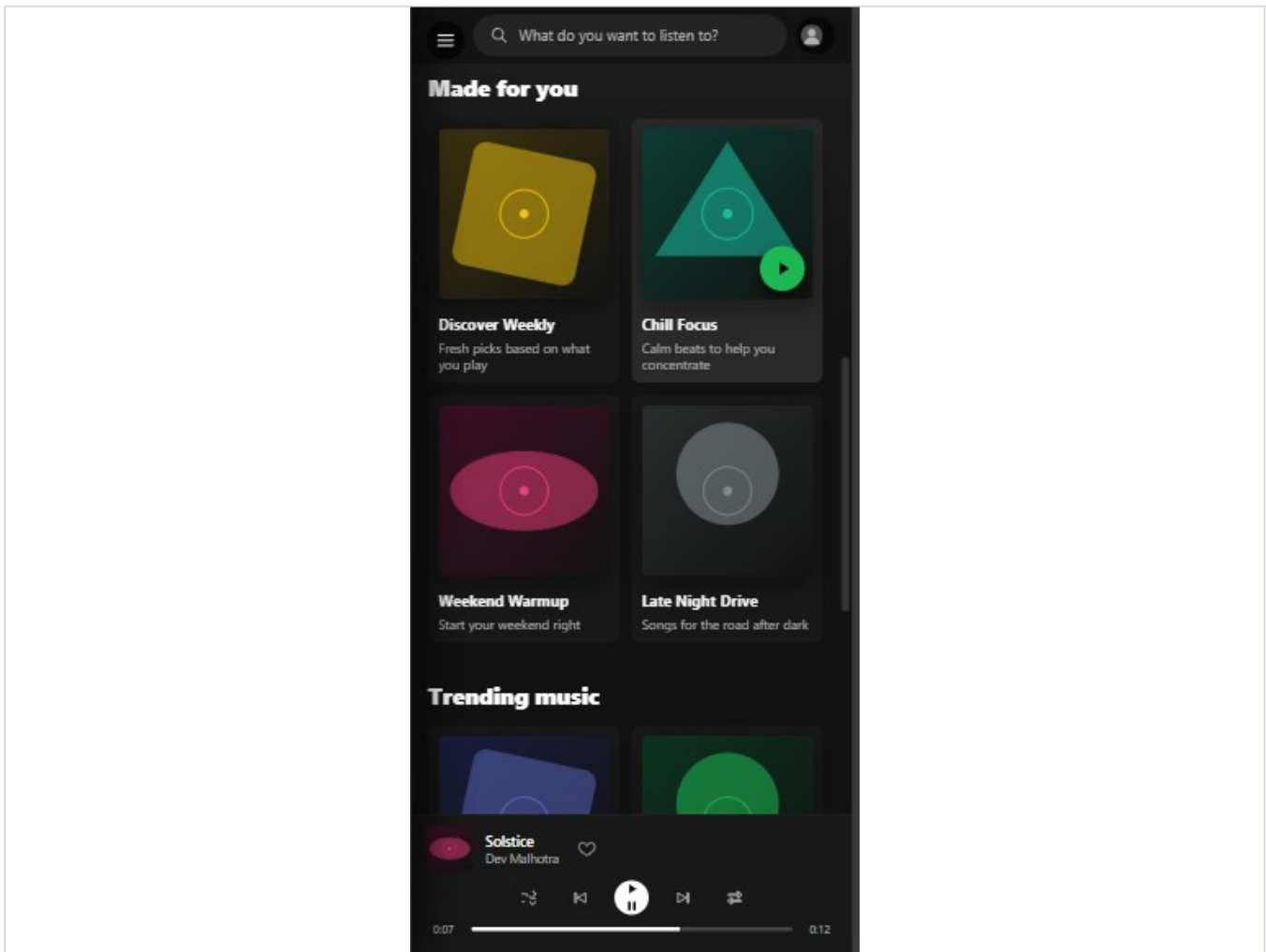
Displays the "Create Playlist" and "Liked Songs" shortcuts alongside the scrollable list of user playlists in the sidebar, including Road Trip Anthems, Study Beats, 90s Throwback, Workout Mix, Rainy Day, and Coffee Shop Jazz.

Figure 4: Navigation / Sidebar — Mobile View



Shows the Groovebox logo, main navigation (Home, Search, Your Library), playlist shortcuts, and the project attribution note, captured at a mobile viewport width. The player bar below shows the track "Solstice" by Dev Malhotra selected.

Figure 5: Interactive Playback — Mobile View



Demonstrates an active playback state on a mobile-width screen: the "Made for you" and "Trending music" card rows are shown in a single-column layout, while the bottom player bar reflects the track "Solstice" playing with the pause icon active and the progress bar at 0:07 of 0:12.

8. Challenges Faced

- Managing UI state in vanilla JavaScript so that the play/pause icon, progress bar, and time display all stay in sync as playback changes, without a framework to manage state automatically.
- Recreating a visually dense, multi-zone layout (sidebar, content grid, fixed player bar) while keeping the player bar reliably anchored to the bottom of the viewport across different content lengths.
- Designing consistent card components for the "Recently played," "Made for you," and "Trending music" sections so that varying content (titles, subtitles, artwork) remained visually uniform.
- Collapsing a multi-column sidebar-plus-grid desktop layout into a clean, single-column mobile experience using CSS responsive techniques, without losing access to navigation or playback controls.
- Handling scrollable sub-sections, such as the sidebar playlist list, independently from the main page scroll.

9. Key Learnings

- Practical experience structuring a complex, multi-zone application layout (sidebar, main content, fixed player bar) using semantic HTML and CSS.
- Stronger UI implementation skills, particularly around recreating card-based content grids and dark-themed interfaces consistent with real-world product design.
- Hands-on JavaScript experience with DOM manipulation and event handling to drive interactive behaviour such as play/pause toggling, progress/time updates, and dynamic track selection.
- Improved understanding of responsive design techniques for collapsing complex, multi-panel desktop layouts into usable mobile interfaces.
- Better project organization practices, including structuring reusable UI components (cards, sidebar items, player controls) for a cleaner, more maintainable codebase.

10. Future Improvements

- Add backend support to persist playlists, liked songs, and playback history beyond the current session.
- Introduce user accounts and authentication so that library and preferences are tied to an individual user.
- Integrate a real music API (subject to licensing) to enable actual audio streaming rather than a UI-only recreation.
- Improve animations and micro-interactions, such as smoother transitions when switching tracks or expanding the player.
- Add search functionality that filters actual playlist/track data as the user types.
- Enhance accessibility with keyboard navigation support and ARIA labelling for all interactive controls.

11. Project Links

Live Demo	https://groovebox-music-player.netlify.app/
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